

PROPOSED SOLUTION ARCHITECTURE

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. System Concept

OptiCrop is an agricultural optimization system that combines machine learning pipelines with an interactive web portal. It runs three models in parallel (Random Forest Classifier, Random Forest Regressor, and K-Means Clustering) to process soil parameters and deliver recommendations to users in real time.

Opticrop - Proposed Solution

2. High-Level System Features

- An AI-driven Agricultural Optimizer delivering five core features:
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- 1. Crop Recommender - RF classifier, 22 crop classes, 100% test accuracy
 - 2. Yield Forecaster - RF regressor, predicts yield in t/ha, 100% test accuracy
 - 3. Soil Health Analyzer - K-Means clustering into 5 soil health clusters
 - 4. NPK Forecaster - RF regressor predicts soil NPK levels
 - 5. Analytics Dashboard - accuracy charts, confusion matrix, feature importance
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